

Experiment 3

Prompt Types for a Support Chatbot

Submission PDF

Source folder on GitHub

Prepared 2026-09-29 from the experiment materials on branch
`codex/complete-experiments`.

Source: Ex03_Prompt_Types_Chatbot/README.md

Ex-3: Prompt Engineering Types - Customer Support Chatbot

Due: 21 Aug 2026 Reference repo: RaajaThilahar/Ex.No.3

Aim

Understand four prompting types by designing a chatbot for **product troubleshooting, order tracking and general inquiries**, then compare AI tools.

Types covered

Straightforward prompt - direct instruction. prompts/1_straightforward.md

Tabular format prompting - ask for / supply a table. prompts/2_tabular.md

Missing-word (cloze) prompting - model fills blanks. prompts/3_missing_word.md

Preceding-question prompting - ask clarifying/prior questions first. prompts/4_preceding_question.md

Workflow

Run every prompt on two tools -> save under outputs/ -> score in evaluation.md. types_explained.md holds the short explanations + examples for the write-up.

Source: Ex03_Prompt_Types_Chatbot/outputs/README.md

Save as <type>_<tool>.md, e.g. tabular_chatgpt.md.

Source: Ex03_Prompt_Types_Chatbot/LAB_RECORD.md

Lab Record: XP-3: Prompt Engineering Types

Field	Entry
Name / Reg. No.	_____
Date	_____
Repo	https://github.com/RHUDHRESH/PROMPT_ENGINEERING-

Aim

Understand straightforward, tabular, missing-word and preceding-question prompting via a support chatbot.

Tools / Apparatus

Claude (saved earlier) and ChatGPT (current session)

Procedure

Prompts written (see prompts/ or the folder README).

Each prompt run on every tool; raw outputs saved under outputs/ (or the folder's equivalent).

Results scored in evaluation.md on: Relevance, tone, structure followed, accuracy.

Findings summarised below.

Observations

See `evaluation.md`. Add your own scores for any tool you ran yourself. Tools not run are left blank; do not fill them in from memory.

Result

Four support prompt types have sample responses for ChatGPT and Claude. Straightforward prompts worked for routine help; preceding questions produced a more careful reply to a late-order complaint.

Conclusion

Use short prompts for simple issues, templates for routine wording, tables for design review and preceding questions when complaints need careful handling. The ratings are provisional and should be reviewed by a person.

Faculty signature: _____

Source: `Ex03_Prompt_Types_Chatbot/evaluation.md`

Evaluation (score 1-5)

ChatGPT responses were prepared in the current Codex session; Claude responses and ratings were present in the repository. Scores are provisional desk reviews, not independent human ratings. Timing was not recorded.

Prompt type	Tool	Relevance	Tone/UX	Structure followed	Accuracy	Notes
Straightforward	ChatGPT	5	5	5	5	Both responses follow the request, give actionable steps and state limits around order lookup.
Straightforward	Claude	5	5	5	4	Existing score; helpful, though some steps and timing assumptions are generic.
Tabular	ChatGPT	5	4	5	4	Six rows with the requested categories and escalation column; policy-dependent answers avoid fixed promises.
Tabular	Claude	5	4	5	4	Existing score; six rows and sensible safety escalation.
Missing word	ChatGPT	5	4	5	4	All blanks completed in the requested pattern; safety check is sensible but inferred beyond supplied details.

Prompt type	Tool	Relevance	Tone/UX	Structure followed	Accuracy	Notes
Missing word	Claude	5	4	5	4	Existing score; wording fits the sentence, with inferred brand and service-time details.
Preceding question	ChatGPT	5	5	5	5	Answers the three lead-in questions and avoids inventing a delivery date or claiming database access.
Preceding question	Claude	5	5	5	4	Existing score; apology, information request and next step are clear.

Conclusion

The preceding-question technique gives more context-sensitive replies to complaints. Straightforward prompts suit simple troubleshooting. Tabular prompts help audit intents and escalation rules, while missing-word prompts control routine wording. The ChatGPT output ratings are self-reviewed, and the Claude ratings came from the existing repository; an independent human should re-score both sets before treating the difference as a model comparison.

Source:

[Ex03_Prompt_Types_Chatbot/outputs/missing_word_chatgpt.md](#)

Missing-word prompt - ChatGPT output

Hi Priya! Thanks for contacting SmartKettle support. I'm sorry your kettle is not heating. Let's fix it: first, unplug it and check that the base is dry and the kettle contains water; if that doesn't work, let it cool and try a different outlet. If the problem continues, I'll connect you with a specialist so they can help within one business day.

Source:

[Ex03_Prompt_Types_Chatbot/outputs/missing_word_claude.md](#)

Missing-word prompt - Claude output

Tool: Claude (answered directly; not ChatGPT).

"Hi Priya! Thanks for contacting SmartKettle support. I'm sorry your SmartKettle X2 is not heating. Let's fix it: first, unplug the kettle, let it cool, and make sure the base and contacts are dry and the kettle is seated firmly, then plug it in again; if that doesn't work, try a different power outlet and descale the kettle if you see limescale build-up. If the problem continues, I'll open a repair or replacement ticket so a specialist can help within one business day."

Notes (not part of the requested output): the brand name "SmartKettle" was inferred from the product name because the context gives no company name, and "one business day" is an illustrative service time, not a real policy.

Source:**Ex03_Prompt_Types_Chatbot/outputs/preceding_question_chatgpt.md**

Preceding-question prompt - ChatGPT output

Q1. Information needed

Order number, contact email, promised delivery date, current tracking status and delivery address confirmation through a secure support channel.

Q2. Likely causes

Carrier delay, an incorrect or incomplete address, or a tracking update that has not synchronized with the carrier scan.

Q3. Reassuring tone

Apologise for the inconvenience, acknowledge the lack of updates, avoid blaming the customer, and give a clear next step without promising an unverified delivery date.

Reply

Im sorry your order is three days late and that you were not kept informed. I cant see your order from this chat, but I can help get it checked. Please contact our support team through the secure order page with your order number and checkout email. They can confirm the carrier status and whether the address needs attention. I wont guess at a new delivery date until it is confirmed.

Source:**Ex03_Prompt_Types_Chatbot/outputs/preceding_question_claude.md**

Preceding-question prompt - Claude output

Tool: Claude (answered directly; not ChatGPT).

Q1. What information is needed to help a customer whose order is late? Order number, the email or name used at checkout, the promised delivery date, and the carrier/tracking number. Also the delivery address, to check for typos.

Q2. Three most likely reasons for a delay?

Carrier or logistics delay (weather, high volume, missed scan).

Stock or fulfilment delay at the warehouse (item back-ordered, processing backlog).

Address or delivery problem (incorrect address, failed delivery attempt, customs for international orders).

Q3. What tone reassures an upset customer? Apologetic but calm, takes ownership ("that's on us for not updating you"), specific rather than vague, and action-oriented with a clear next step.

Reply to: "My order was due 3 days ago and nobody told me anything!"

I'm really sorry. Your order should have arrived by now, and you should have heard from us before you had to ask. That's on us, and I understand why you're frustrated.

Here's what I'll do right away: I'll look into where your parcel is with the carrier. To do that, could you send me your order number and the email address you used at checkout? It also helps to know the delivery date you were given and the address on the order, so we can rule out a typo.

Common causes are a carrier delay, a delay at our warehouse, or a delivery problem such as an address issue. Once I know which it is, I'll tell you exactly what happens next, whether that's a new delivery date, a re-shipment, or a refund, and I'll keep you updated so you're not left wondering again.

Source:

[Ex03_Prompt_Types_Chatbot/outputs/straightforward_chatgpt.md](#)

Straightforward prompt - ChatGPT output

Tool: ChatGPT in the current Codex session. Exact model version and response time were not exposed or measured.

Prompt A - Headphones will not connect

Sorry about that. Try these steps:

Charge the headphones and turn Bluetooth off and on on your phone.

Put the headphones in pairing mode using their instructions.

On your phone, remove the old headphone pairing, then search and pair again.

Restart both devices and try once more.

If that does not work, tell me the headphone model and phone model so I can suggest the next checks. Stop if the headphones become hot or smell unusual.

Prompt B - Order status without database access

Im sorry your order is causing worry. I cant access order records here, so I cant look up #48213. Please use the tracking link in your shipping email or open your accounts order history. If you share the email address used at checkout through the stores secure support channel, a staff member can investigate. Dont post payment details in chat.

Source:

[Ex03_Prompt_Types_Chatbot/outputs/straightforward_claude.md](#)

Straightforward prompt - Claude output

Tool: Claude (answered directly by the model writing this file; not ChatGPT).

Prompt A - headphones won't connect (max 120 words)

Sorry about that, let's get them connected! Try these steps:

Switch your phone's Bluetooth off and on again.

Put the headphones in pairing mode (usually hold the power button until the light flashes).

In your phone's Bluetooth settings, "forget" any old entry for the headphones, then pair them again.

Make sure they have some charge, then restart both the headphones and the phone.

If they still won't connect, tell me the headphone model and your phone type and I'll dig deeper. I can also arrange a repair or replacement if needed.

(Word count of reply: about 100, within the 120 limit.)

Prompt B - "Where is my order #48213?" (no database access)

Hi there, thanks for getting in touch, and I'm sorry for the worry. I can't look up orders from this chat, so I can't see where #48213 is right now. Here is the quickest way to find out:

- Open the shipping-confirmation email we sent you. It contains a tracking number and a "Track package" link.
- Or log in to your account, go to "My orders", and open order #48213 for live status.
- If you can't find the email, check your spam folder, or send me the email address used at checkout and I'll point you to the right place.

If tracking has not updated in 3 business days, reply here or contact our order team with your order number and they can investigate with the carrier.

Source: [Ex03_Prompt_Types_Chatbot/outputs/tabular_chatgpt.md](#)

Tabular prompt - ChatGPT output

Issue category	Example customer message	Bot reply (max 25 words)	Escalate to human?
Troubleshooting	My headphones will not pair.	Charge them, enable pairing mode, remove the old phone pairing, then reconnect. Share both device models if it still fails.	N
Troubleshooting	My kettle will not heat.	Unplug it, check the base and water level, then retry. Stop using it if damaged, leaking or smelling burnt.	Y
Order tracking	Where is my parcel?	Use the carrier link in your shipping email or order history. If tracking is missing, contact support with your order number.	N
Order tracking	My parcel is marked delivered but missing.	Check the delivery photo and nearby safe places. If it remains missing, contact support so a person can investigate with the carrier.	Y
General inquiry	What is your return policy?	I can help find it. Which country did you order from, and was the item bought online or in-store?	N
General inquiry	Can I change my delivery address?	Contact support immediately with your order number. Address changes depend on dispatch status, so I cannot promise the carrier can redirect it.	Y

Source: [Ex03_Prompt_Types_Chatbot/outputs/tabular_claude.md](#)

Tabular format prompt - Claude output

Tool: Claude (answered directly; not ChatGPT).

Issue category	Example customer message	Bot reply (max 25 words)	Escalate to human? (Y/N)
Product troubleshooting	"My wireless earbuds won't pair with my phone."	Sorry about that! Put the earbuds in their case, hold the button until the light flashes, then forget and re-pair them in Bluetooth settings.	N

Issue category	Example customer message	Bot reply (max 25 words)	Escalate to human? (Y/N)
Product troubleshooting	"My smart kettle stopped heating and smells burnt."	Please unplug it now and stop using it. A burning smell can be a safety issue, so I'm connecting you to a specialist.	Y
Order tracking	"Where is my order #48213?"	I can't view orders here, but your shipping email has a tracking link, or check My Orders in your account.	N
Order tracking	"Tracking says delivered but I never got it."	I'm sorry! Please check with neighbours and your porch first; if it's still missing, I'll pass this to our team to investigate.	Y
General inquiry	"What is your return policy?"	You can usually return unused items within the stated return window. Check our Returns page for exact days and steps.	N
General inquiry	"Do you ship internationally?"	Shipping options depend on your country. Enter your address at checkout to see available delivery methods and costs.	N

Notes on the table: the two "Y" rows are safety (burning smell) and a possible lost parcel, where a human must act. The return-window and shipping answers deliberately avoid inventing store-specific numbers (illustrative wording only; a real store would insert its policy). Reply word counts were kept at or below 25.

Source: [Ex03_Prompt_Types_Chatbot/prompts/1_straightforward.md](#)

Straightforward

You are a friendly customer-support chatbot for an electronics store. A customer writes: "My wireless headphones won't connect to my phone." Give clear step-by-step troubleshooting in a conversational tone, max 120 words.

A customer asks: "Where is my order #48213?" You cannot access a database. Reply politely, explain what you need, and tell them how to track it.

Source: [Ex03_Prompt_Types_Chatbot/prompts/2_tabular.md](#)

Tabular format

Act as a support chatbot. Return a table with columns: Issue category | Example customer message | Bot reply (max 25 words) | Escalate to human? (Y/N).

Rows: product troubleshooting (2), order tracking (2), general inquiry (2).

Source: [Ex03_Prompt_Types_Chatbot/prompts/3_missing_word.md](#)

Missing word

Fill in every blank (___) to complete the support chatbot reply. Keep the sentence structure, friendly tone, return only the completed text.

"Hi ___! Thanks for contacting ___ support. I'm sorry your ___ is ___. Let's fix it: first, ___; if that doesn't work, ___. If the problem continues, I'll ___ so a specialist can help within ___."

Context: customer Priya, product SmartKettle X2, issue: won't heat.

Source:

[Ex03_Prompt_Types_Chatbot/prompts/4_preceding_question.md](#)

Preceding question

Before answering, work through these questions in order (show short answers):

Q1. What information is needed to help a customer whose order is late?

Q2. What are the three most likely reasons for a delay?

Q3. What tone reassures an upset customer?

Now, using Q1-Q3, reply to: "My order was due 3 days ago and nobody told me anything!"

Source: [Ex03_Prompt_Types_Chatbot/types_explained.md](#)

Prompt types explained

Type	Idea	When it shines	Risk
Straightforward	One clear instruction, no scaffolding	Simple, well-known tasks	Vague output if the task is ambiguous
Tabular format	Force rows/columns for input or output	Comparisons, checklists, structured data	Model may pad cells or break the table
Missing word	Give a template with blanks to fill	Controlling wording/tone, templates, quizzes	Model may rewrite the template
Preceding question	First question(s) build context, then the real ask	Multi-step reasoning, ambiguous user requests	Longer, more turns

Chatbot examples

- **Straightforward:** "A customer says their router keeps disconnecting. Give troubleshooting steps."
- **Tabular:** "Table: Symptom | Likely cause | Fix | Escalate? for 5 router issues."
- **Missing word:** "Hi ____, your order ____ is currently ____ and should arrive by ____."
- **Preceding question:** "Q1: What info do you need to track an order? Q2: What if the ID is invalid? Now answer: 'Where is my order?'"